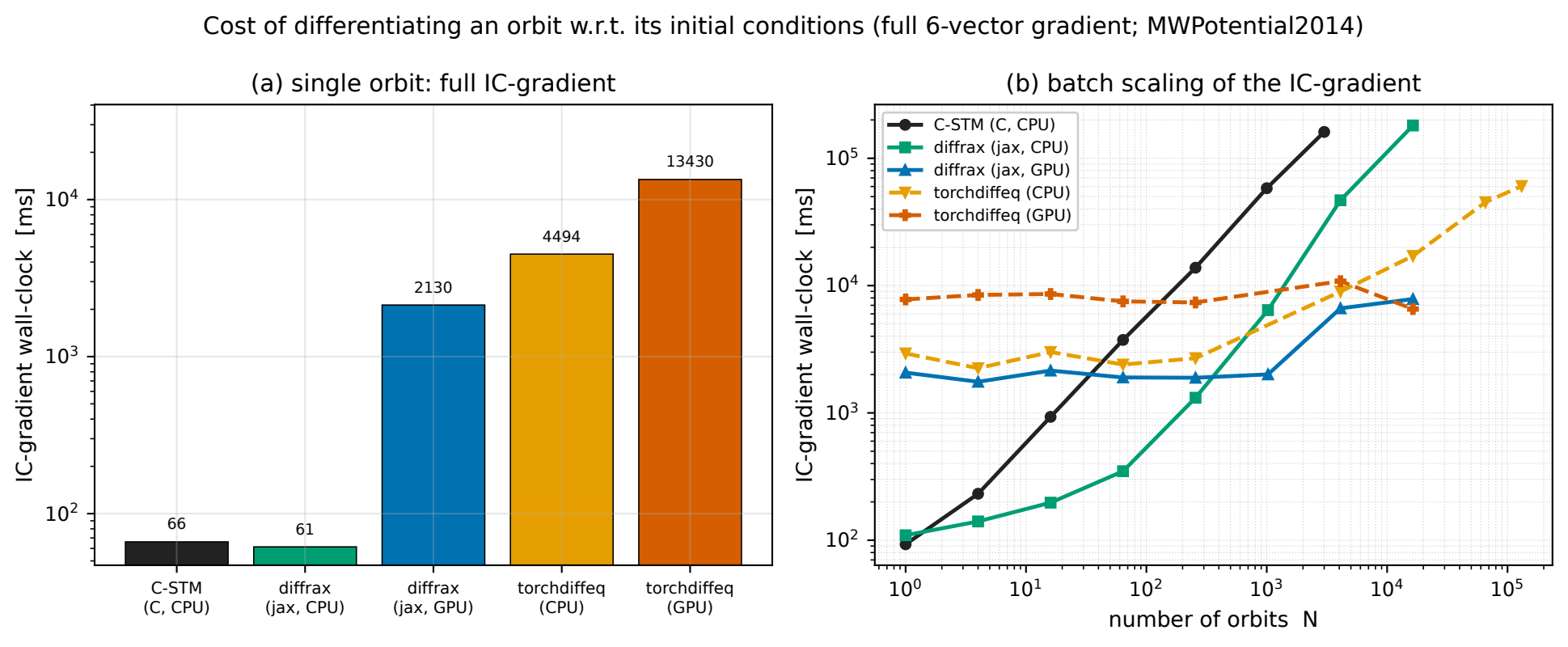
Cost of differentiating an orbit w.r.t. its initial conditions (full 6-vector gradient; MWPotential2014)
(a) single orbit: full IC-gradient
IC-gradient wall-clock [ms]
66
61
2130
4494
13430
C-STM (C, CPU)
diffrax (jax, CPU)
diffrax (jax, GPU)
torchdiffeq (CPU)
torchdiffeq (GPU)
(b) batch scaling of the IC-gradient
C-STM (C, CPU)
diffrax (jax, CPU)
diffrax (jax, GPU)
torchdiffeq (CPU)
torchdiffeq (GPU)
IC-gradient wall-clock [ms]
number of orbits N